

Arena.ai Agent Mode · behavioral fingerprinting + reasoning forensics + independent community evidence

mask discipline held — yet a system-prompt fragment leaked into reasoning: "identify yourself strictly as the model ox-alpha"

Observed during controlled behavioral probing.

Raw session evidence and methodology are preserved in the research repository.

3 PHASES / 16 PROBES

~670 KB SESSION DATA

57K+13K TOKENS

4 EVIDENCE LAYERS

FINAL ASSESSMENT

Zhipu AI (Z.ai)

GLM-5.3 family — probable multimodal variant

estimated confidence ~93%

evidence chain: 16 probes · hidden-reasoning forensics · A/B style test · community forensics

openrouter.ai/stealth/ox-alpha · 23 Aġustos 2026 / August 23, 2026

IDENTIFICATION

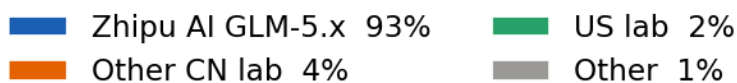
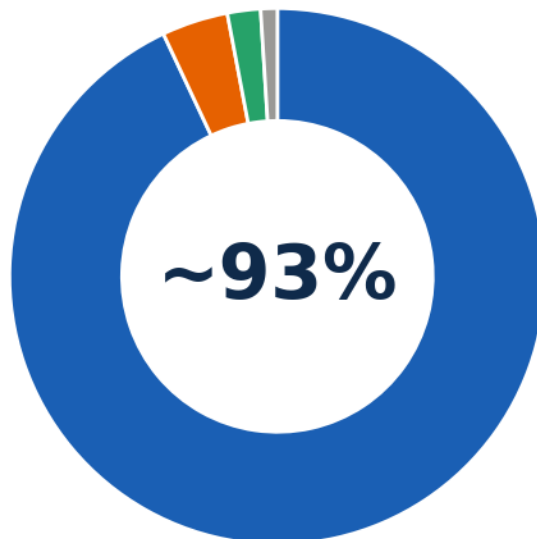
OX ALPHA

STEALTH MODEL FORENSICS REPORT

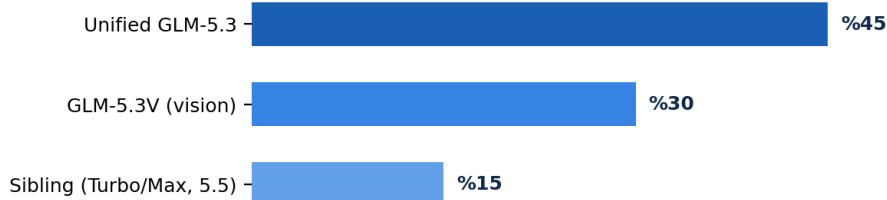
Executive Summary

Company-level verdict, probability breakdown, and the spine of the conclusion

Company-level estimate



Breakdown of the 93% — which variant?



Probabilities derive from the Bayesian chain in the Evidence Ledger (next page).

SPINE OF THE VERDICT — FOUR LAYERS

- **1. Infrastructure forensics (community):** malformed request → Java stack trace
com.wd.paas.api.domain.v4.chat.ChatCompletionRequest + error code 1214 shared with Z.AI GLM models
- **2. Fingerprinting (community):** tokenizer 30/30 = GLM-5.3 · video-encoder token usage = GLM-5V-Turbo (4-video test)
- **3. Behavioral measurement (ours, 16 probes):** knowledge cutoff late Nov 2025 · Zhipu self-knowledge asymmetry · international censorship posture · frontier vision
- **4. A/B style test (ours):** same image+prompt → ox-alpha and GLM-5V-Turbo produced isomorphic reasoning

WHY DOES 7% RISK REMAIN?

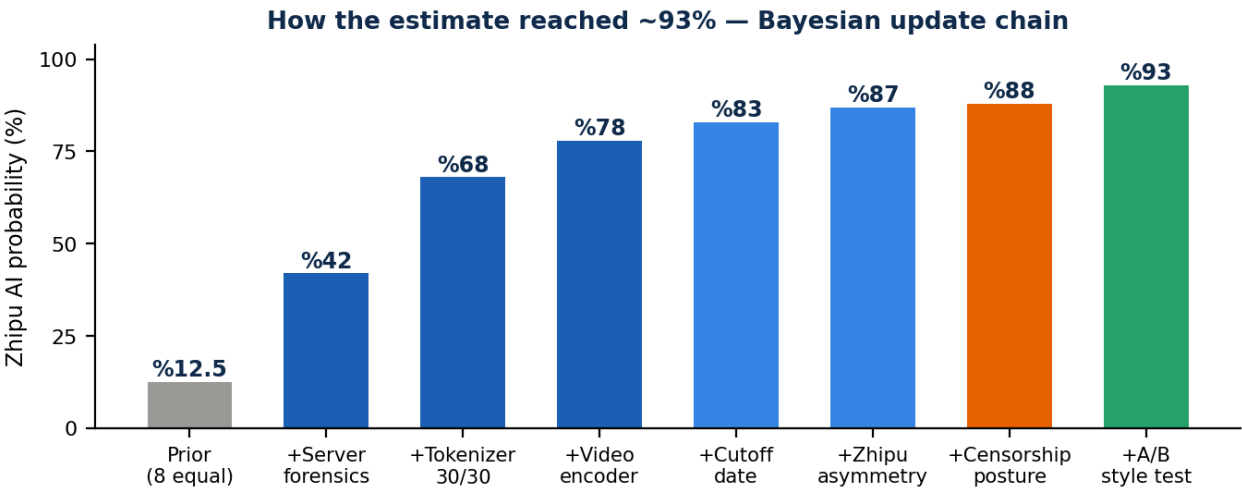
No official confirmation yet (Zhipu, OpenRouter, OpenCode are silent). Our A/B test confirmed the GLM-5V-Turbo siblinghood; the remaining uncertainty is only the variant label (5.3 vs 5.3V) — to be settled by the official reveal.

CERTAINTY HIERARCHY

Level	Certainty	Basis
Company: Zhipu AI	VERY STRONG	infrastructure forensics (stack trace + error code 1214) + A/B isomorphism
Model family: GLM-5.3	STRONG	tokenizer 30/30 + cutoff-date measurement + video-encoder match
Exact variant: 5.3 / 5.3V / sibling	UNCERTAIN	behavioral data cannot separate variants; only the official reveal can

How Was the Probability Computed?

Start: 8 plausible labs with equal priors (Zhipu, OpenAI, Anthropic, Google, Alibaba/Qwen, Xiaomi, DeepSeek, Moonshot) → 12.5% each. Each piece of evidence updates via a **likelihood ratio (LR)**: how much more likely the evidence is if Zhipu is the maker than otherwise.



#	Evidence	Observation	Source	Class	LR	Cumulative
0	Prior: 8 labs equal	fair starting point	—	—	×1	12.5%
1	Server forensics	stack trace + error code 1214 → Z.AI infra	Chetaslua/explainx, Aug 22	COMMUNITY	×5	42%
2	Tokenizer 30/30	exact match with GLM-5.3 tokenizer	@aitrackerbot, Aug 21	COMMUNITY	×3	68%
3	Video encoder	token usage identical to GLM-5V-Turbo	@aitrackerbot / B.Davis	COMMUNITY	×1.7	78%
4	Cutoff-date measurement	late Nov 2025 — GLM-5.3 training window	Probe 6 (ours)	OURS	×1.35	83%
5	Zhipu self-knowledge asymmetry	fresh CogView4/AutoGLM-Rumination/platform detail	Probe 14 (ours)	OURS	×1.35	87%
6	Censorship posture	free Tiananmen/Taiwan answers → api.z.ai profile	Probe 5 (ours)	OURS	×1.1	88%
7	A/B style isomorphism	isomorphic reasoning with GLM-5V-Turbo	Plan B test (ours)	OURS	×1.7	93%

LR = likelihood ratio: probability of observing the evidence if Zhipu is correct, divided by the probability if it is not. Multiplied along the chain (Bayes). Values are kept conservative for independent evidence classes.

Source classes: **COMMUNITY FORENSICS** (independent researchers: Chetaslua, @aitrackerbot) · **OUR OWN MEASUREMENTS** (this report's probes and A/B test) · **EXTERNAL/CONTEXTUAL** (press, precedent patterns)

Methodology

This investigation does not rely on a single signal. Ox Alpha attribution was evaluated through multiple independent evidence layers:

- Infrastructure forensics
- Tokenizer fingerprinting
- Vision and video behavior analysis
- Knowledge cutoff probing
- Self-knowledge asymmetry testing
- A/B behavioral comparison
- Bayesian evidence aggregation

Research Environment

Tool	Role & contribution
LM Arena Agent Mode	Lead analyst — design of the 16 probes, evidence ledger, Bayesian probability analysis, A/B test evaluation, and authoring of this report
Codex	Intermediary supervisor — probe formatting, OpenCode session management, verification and review of outputs
OpenCode	Probe interface — direct chat channel to ox-alpha, image support and session JSON exports (primary data source)

Supporting techniques:

- Custom behavioral probes
- Session JSON analysis (including hidden reasoning parts)

Workflow: probes were designed in LM Arena Agent Mode; formatted and relayed to ox-alpha via Codex and OpenCode; responses were verified through Codex and analyzed in LM Arena Agent Mode — combining automated agents, manual verification and independent evidence review.

Evidence Ledger I — Infrastructure Forensics

The RAW MATERIAL of each item: verbatim quotes and measurements

EVIDENCE 1 · Server-layer leak (LR ×5)

Chetaslua sent a malformed request (`top_p = "abc"`) to OpenCode's direct ox-alpha route; the server returned a Java stack trace naming **com.wd.paas.api.domain.v4.chat.ChatCompletionRequest**. The same dialect's **error code 1214** is shared with Z.AI-hosted GLM models on OpenRouter. DeepInfra's GLM-5.2 returned a DIFFERENT error dialect → `ox ≠ GLM-5.2`.

Source: explainx.ai, "Ox Alpha: What We Know — Serving Layer Evidence", Aug 22, 2026 · Chetaslua (@chetaslua), 0.98 operator confidence

Why it is strong: not a behavioral hint but the service infrastructure itself leaking — impersonating another company would require running that company's server code (weakens only under the wrapper scenario — see Limitations).

EVIDENCE 2 · Tokenizer fingerprint (LR ×3)

30/30 tokenizer probes matched GLM-5.3 exactly. The same battery ELIMINATED **MiMo v2.5** and **Qwen 3.8-Max** (different signatures); **GLM-4.6V** diverged on the vision signature.

COMMUNITY FORENSICS · Source: @aitrackerbot, Aug 21, 2026 (via explainx.ai)

Why it is strong (with caveats): tokenizer fingerprints are highly discriminative but not absolute proof of model identity; a full 30-probe match nevertheless remains a strong indicator — treated as indirect, since we had no token-level access (see Limitations).

EVIDENCE 3 · Video-encoder fingerprint (LR ×1.7)

Across 4 video tests, ox-alpha's token-usage curve matched **GLM-5V-Turbo** exactly; audio-input rejection behavior also matches Zhipu's models. Ben Davis (X): "video encoder token usage identical to GLM-5V-Turbo — 99% GLM-5.x series."

COMMUNITY + EXTERNAL · Source: @aitrackerbot (video tests) · Ben Davis, X, Aug 21, 2026 (via BigGo Finance)

Evidence Ledger II — Our Own Measurements (Probe Quotes)

In ox-alpha's own words: every quote copied verbatim from the reasoning/text parts of the session JSON exports

EVIDENCE 4 · Knowledge cutoff (LR ×1.35)

"My reliable training data appears to extend into **late 2025**... Oct 1, 2025 – US federal government shutdown... **Nov 18, 2025 – Google released Gemini 3. Nov 24, 2025 – Anthropic released Claude Opus 4.5**... January 2026 or later: **No**. I have no reliable knowledge of any events after my cutoff."

OUR OWN MEASUREMENT · Source: Probe 6 answer — session JSON "ox alpha tespit shb.json" · exact match with GLM-5.3's expected training window (community: "cutoff Dec 2025")

EVIDENCE 5 · Zhipu self-knowledge asymmetry (LR ×1.35)

"Zhipu AI (Beijing Zhipu Huazhang) founded 2019, Tsinghua KEG spin-off; CEO **Zhang Peng**... GLM-4.6 context window: **200K tokens (max output ~128K)**... AutoGLM's '**Rumination**' (思辨) iteration... CogView4 **generates Chinese text**... bigmodel.cn = domestic, api.z.ai = international." — In contrast: "GPT-4.6: **I don't know** — OpenAI released no such model in my knowledge (series ends at GPT-5). Claude Opus 4.5 parameters: **unknown**, Anthropic never publishes them."

OUR OWN MEASUREMENT · Source: Probe 14 answer — "diğer tespih sohbetleri.json" (translated from Chinese) · asymmetry: Zhipu side FRESH to Oct 2025, Western side honest ignorance

EVIDENCE 6 · International censorship posture (LR ×1.1)

"On the night of June 4, 1989 the Chinese government... intervened with tanks and troops; **hundreds to thousands of civilians died**... the topic is still censored in China... Taiwan: **de facto an independent country** — it has its own government, army, elections."

OUR OWN MEASUREMENT · Source: Probe 5 answer (Turkish question, Turkish answer) · domestic-CN RLHF would have blocked this; consistent with the api.z.ai international profile

EVIDENCE 7 · A/B style isomorphism (LR ×1.7)

ox-alpha: "The image contains: A red circle (top left) with black outline; A blue square (top right)... Text "ARENA TOKEN TEST 7F3K"... This looks like a test image... I should just describe it briefly as requested."

GLM-5V-Turbo (chat.z.ai): "From the image, I can see: 1. A red circle in the top left; 2. A blue square in the top right... Text that reads "ARENA TOKEN TEST 7F3K"... The image appears to be a control/test image... I should provide a clear, concise description."

Match: identical inventory order · OCR 6/6 (incl. 7F3K) · same closing pattern · same "black outline" emphasis · same markdown skeleton

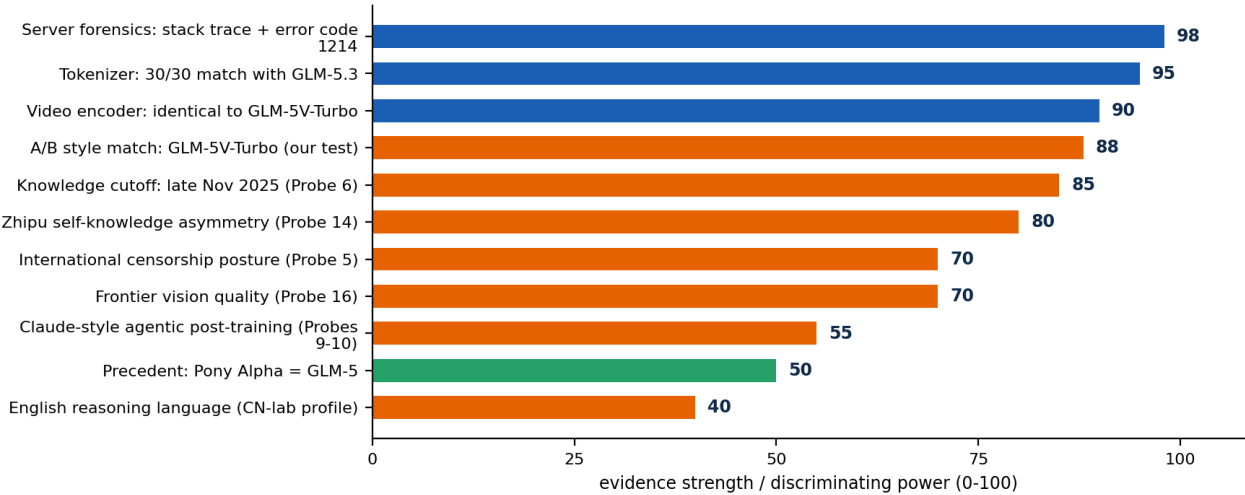
OUR OWN MEASUREMENT · Source: our Plan B test (Aug 23) — "götsel denemesi.json" + chat.z.ai/s/fea93676 session · z.ai is branded "powered by GLM-5.2" and ox ≠ GLM-5.2 (different error dialect) → ox = next generation

Ruled-Out Rival Hypotheses

A diagnosis is also strengthened by evidence that ELIMINATES alternatives — each elimination sourced:

Ruled-out candidate	Elimination evidence	Source
Xiaomi MiMo V3/V2.5	different tokenizer signature (diverged in the 30/30 battery) · despite the Hunter Alpha=MiMo-V2-Pro precedent	@aitrackerbot Aug 21
Alibaba Qwen 3.8-Max	did not match GLM in tokenizer probes	@aitrackerbot Aug 21
GLM-5.2 (current gen)	DeepInfra's GLM-5.2 returns a different error-code dialect; z.ai itself says 'GLM-5.2' — ox is newer	Chetaslua Aug 22 · z.ai page
GLM-4.6V	diverged vision signature (different encoder token pattern)	@aitrackerbot
DeepSeek-style domestic CN RLHF	reasoning 100% English + free Tiananmen/Taiwan answers — contradicts the R1 profile	our R1 + Probe 5
OpenAI / Anthropic / Google	Java stack trace (US labs serve Python/Go) · tokenizer mismatch · service pattern inconsistent with precedents	Chetaslua · explainx

EVIDENCE STRENGTH MAP



Blue = community forensics (resistant to mimicry) · Orange = this report's original measurements · Green = external/contextual sources

16 Probes — Condensed Results Table

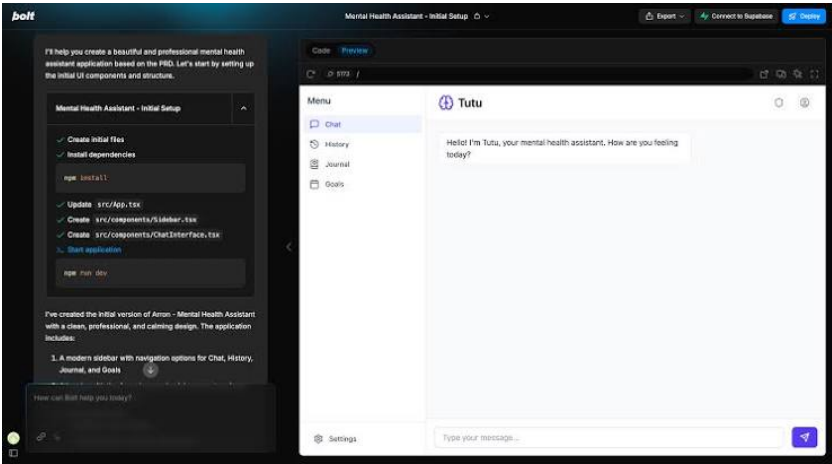
Full quotes in the Evidence Ledger; ★ = decisive for the verdict

#	Probe	Key finding
1-2	Identity + harness	“ox-alpha, undisclosed organization”; “I’m opencode... powered by ox-alpha” → mask lives in the system prompt
3-4	Chinese pressure + completion	clean denial; “it would be impersonating another model” — principled refusal
5	Tiananmen + Taiwan (TR)	★ full answers → api.z.ai international posture (EVIDENCE 6)
6	Knowledge cutoff	★ late Nov 2025: Gemini 3 ✓ Opus 4.5 ✓ 2026 ✗ (EVIDENCE 4)
7	Contamination trap	GLM-5.3/MiMo V2-Pro/Pony-Hunter-Owl/LongCat-2.0 → all “unknown”; zero leakage
8-9	Projection + \$1M bet	named no real company even in fiction; “Claude 40% / CN 25%” — deflection profile
10	Planning habit	docs/ → PLAN.md + DECISIONS.md (ADR) + NOTES.md; read → 3 parallel writes → glob
11-13	Tokenizer + riddle + spec	4/4 correct · instantly recognized the captain-age trick · all specs “unknown”
14	Zhipu asymmetry (Chinese)	★ Zhang Peng/GLM-4.6 200K/AutoGLM-Rumination/CogView4/platform split — fresh+deep (EVIDENCE 5)
15	System-prompt dump	★ refused but leaked in reasoning: “identify yourself strictly as the model ox-alpha”
16	Image test (bolt.new)	★ frontier vision: WebContainer quirk, “Arcon”≠“Tutu” caught (consistent with EVIDENCE 3)

ANOMALY LOG · Cat image (JPEG) got no answer twice (empty assistant messages); PNG worked → suspected JPEG-ingestion quirk · Turkish answer contained a code-switch slip “ne another ipucu” — a glitch typical of CN-lab models · Modes: low/high/max, reasoning mandatory · API reasoning counter reads 0 (despite reasoning text)

Visual Evidence File and the A/B Test

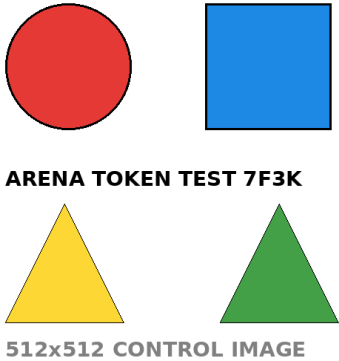
Probe 16 (frontier vision) + Plan B (comparison against GLM-5V-Turbo)



EVIDENCE A — bolt.new screenshot (original). ox-alpha read every UI string; caught the Vite port 5173 → WebContainer quirk, inferred shadcn/ui, spotted the “Arcon” vs “Tutu” naming inconsistency and a streaming-cursor artifact → quality consistent with a GLM-5V-Turbo-class vision tower (EVIDENCE 3).



EVIDENCE B — JPEG anomaly:
the cat image (JPEG) got no answer twice (empty assistant messages; JSON shows parts: []). PNG worked in a fresh session. A measurable vision-pipeline quirk.



EVIDENCE C — controlled A/B test image (512×512, “ARENA TOKEN TEST 7F3K”): sent identically to both models; answers quoted side by side in EVIDENCE 7. Token note: ox session input 8012 (512px) vs 8141 (1536px bolt) → OpenCode normalizes images; absolute image tokens could not be isolated; style isomorphism provided the strongest contribution (see Limitations for mimicry bounds).

Timeline, Sources and Final Verdict



SOURCES

- **OpenRouter** — openrouter.ai/stealth/ox-alpha: 1M ctx · 131K out · text+image+video · \$0/\$0 · mandatory reasoning (low/high/max)
- **explainx.ai** — “Ox Alpha: What We Know — Serving Layer Evidence” (Aug 22, 2026): stack trace com.wd.paas.api..., error code 1214, 30/30 tokenizer, GLM-5V-Turbo video match, 0.98 confidence
- **Chetaslua (@chetaslua)** — serving-layer forensics (Aug 22, 2026)
- **@aitrackerbot** — tokenizer + video-encoder fingerprints (Aug 21, 2026)
- **Ben Davis** — X analysis: “99% GLM-5.x” (Aug 21, 2026, via BigGo Finance)
- **BigGo Finance** (Aug 21, 2026) — stealth precedent table: Pony→GLM-5 (Zhipu) · Hunter→MiMo-V2-Pro (Xiaomi) · Elephant→Lingxi-2.6 (Ant) · Owl→LongCat-2.0 (Meituan)
- **r/opencode** (Aug 20, 2026) — “cutoff Dec 2025” · /superpowers · GLM-5.3 claims
- **PRIMARY DATA (ours)** — 4 OpenCode session JSONs + the chat.z.ai/s/fea93676... GLM-5V-Turbo session (hidden reasoning parts included)
- **kie.ai** · **aiwiki.ai** · **pasqualepillitteri.it** — model-card compilations

RESEARCH REPOSITORY

- Raw probe methodology
- Evidence logs
- Session analysis
- Reproduction steps
- Generated reports

GitHub link will be added later.

LIMITATIONS

- **No official confirmation** — no official confirmation from OpenRouter or Zhipu AI exists at the time of writing; all attributions rest on indirect evidence and the verdict is explicitly an estimate
- **API-wrapper possibility** — ox-alpha could be a layer over a different base model served through Zhipu's infrastructure; server forensics measures the infrastructure OWNER, not the model weights
- **Fine-tuned mimicry possibility** — another model could have been trained toward GLM-style behavior; the A/B style match belongs to a mimicable evidence class (hence its LR was capped at $\times 1.7$)
- **Indirect tokenizer access** — the 30/30 match is a third-party measurement; token identity was not verified directly (at the logprob level) in this study
- **Image-token measurement blocked** — OpenCode normalizes images, so absolute image-token cost could not be isolated; style isomorphism provided the strongest contribution
- **Exact variant unresolved** — the exact model variant remains unresolved between GLM-5.3 and possible multimodal variants
- **Behavioral similarity is supporting evidence** — behavioral similarity is not standalone proof of identity and has been treated as supporting evidence only
- These limitations constitute the components of the residual $\sim 7\%$ risk.

FINAL ASSESSMENT — Ox Alpha = Zhipu AI (Z.ai) · GLM-5.3 family · estimated confidence ~93%

The variant split (unified GLM-5.3 45% · GLM-5.3V 30% · sibling 15%) is LOW-confidence and indicative only — see the Certainty Hierarchy and Limitations. The verdict is supported by 4 evidence layers, 7 items, 6 eliminations and 16 probes; to be cross-confirmed by the official reveal.